

INDICES (Powers)

An Index (plural: Indices), also known as an exponent or power, is the small number that indicates how many times a base number is multiplied by itself.

$$5^3 = 5 \times 5 \times 5$$

$$7^4 = 7 \times 7 \times 7 \times 7$$

Identities

$$1) \quad a^m \times a^n = a^{(m+n)}$$

$$5^5 \times 5^8 = 5^{(5+8)} = 5^{13}$$

$$11^2 \times 11^7 = 11^{(2+7)} = 11^9$$

$$13^5 \times 13^{12} = 13^{(5+12)} = 13^{17}$$

Conditions :-

- Base should be same
 - Multiplication "x" sign
- $\Rightarrow$ then, powers add Hongi

$$2) \quad a^m \div a^n = a^{(m-n)}$$

$$5^8 \div 5^5 = 5^{(8-5)} = 5^3$$

$$11^{11} \div 11^7 = 11^{(11-7)} = 11^4$$

$$12^8 \div 12^3 = 12^{(8-3)} = 12^5$$

$$105^8 \div 105^4 = 105^{(8-4)} = 105^4$$

Conditions :-

- Base should be same
- Division "÷" sign

$\Rightarrow$ powers subtract Hongi

$$\cdot \frac{2^6}{2^2} = \frac{(64)}{(4)} = 16 = 2^4 \quad \{ \text{Hence, } 2^{(6-2)} = 2^4 \}$$

$$\cdot \frac{5^5}{5^4} = \frac{3125}{625} = 5 = 5^1 \quad \{ \text{Hence, } 5^{(5-4)} = 5^1 \}$$

$$\cdot \frac{3^5}{3^3} = \frac{243}{27} = 9 = 3^2 \quad \{ \text{Hence, } 3^{(5-3)} = 3^2 \}$$

$$3) (a^m)^n = a^{mn}$$

{ Base should be same }

$$(6^2)^3 = 6^{(2 \times 3)} = 6^6$$

$$(2^3)^2 = 2^{(3 \times 2)} = 2^6$$

$$\cdot 8^2 = 64$$

↓

$$[2^3]^2 = 2^{(3 \times 2)} = 2^6 = 64.$$

$$\cdot 9^3 = 729$$

↓

$$[3^2]^3 = 3^{(2 \times 3)} = 3^6 = 729.$$

$$4) (ab)^n = a^n b^n$$

{ Powers same and base different }

$$(4 \times 5)^2 = 4^2 \times 5^2$$

$$\cdot (35)^2 = (7 \times 5)^2 = 7^2 \times 5^2$$

$$\cdot (64)^2 = (16 \times 4)^2 = 16^2 \times 4^2$$

$$5) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

{ Powers same and base different }

$$\cdot \left(\frac{2}{7}\right)^5 = \frac{2^5}{7^5}$$

$$\cdot \left(\frac{25^2}{5^2}\right) = \frac{625}{25} = 25 = 5^2$$

and

$$\frac{25^2}{5^2} = \left(\frac{25}{5}\right)^2 = (5)^2$$

$$6) a^0 = 1 \quad (a \neq 0)$$

$$18^0 = 1$$

$$17^0 = 1$$

$$4^0 = 1$$

$$x^0 = 1$$

$$k^0 = 1$$

$$m^0 = 1$$

Any value with power [0, zero]
will be equal to 1.

$$a^0 = a^{n-n}$$

$$= \frac{a^n}{a^n} \Rightarrow 1.$$

$$[\text{Hence, } a^0 = 1]$$

$$\cdot (5^2 \times 3^2)^0 = 1$$

$$\cdot (6^3 \times 5^2 \div 3^6)^0 \Rightarrow \left(\frac{6^3 \times 5^2}{3^6} \right)^0 = 1.$$

$$7) a^{-n} = \frac{1}{a^n}$$

$$15^{-6} = \frac{1}{15^6}$$

$$23^{-6} = \frac{1}{23^6}$$

$$(2^2 \times 5)^{-2} = \frac{1}{(2^2 \times 5)^2}$$

{ During Reciprocal,
power ka sign change ho jayega }

$$a^{-n} = a^{0-n} = \frac{a^0}{a^n} = \frac{1}{a^n}$$

$$a^n = a^{0-(-n)} = \frac{a^0}{a^{-n}} = \frac{1}{a^{-n}}$$

$$3^{-100} = \frac{1}{3^{100}}$$

$$3^{-17} = \frac{1}{3^{17}}$$

$$17^{-89} = \frac{1}{17^{89}}$$

Also, Reverse Case

$$3^7 = \frac{1}{3^{-7}}$$

$$14^5 = \frac{1}{14^{-5}}$$

$$17^{101} = \frac{1}{17^{-101}}$$

$$3^5 \times 9 \times 3^2 = 3^?$$

- (A) 8 (B) 7 (C) 5 ☒ (D) 9 (E) None of these

$$3^5 \times 3^2 \times 3^2 \Rightarrow 3^{(5+2+2)} = 3^9$$

$$3^9 = 3^x$$

$$x = 9$$

[Base same
Multiplication sign
So, powers add]

$$(6^5 \div 6^2) \div 6^1 = ?$$

- (A) 6 ☒ (B) 36 (C) 18 (D) 12 (E) 72

$$(6^5 \div 6^2) \div 6^1$$

$$6^{(5-2)} \div 6^1 \Rightarrow 6^3 \div 6^1 \Rightarrow 6^{(3-1)} \Rightarrow 6^2$$

$$6^2 = 36$$

[Base same
Division sign
So, powers subtract]

$$(49^2 \times 7^3) \div 343^2 = ?$$

- (A) 1 ☒ (B) 7 (C) 49 (D) 343 (E) 2401

$$[(7^2)^2 \times 7^3] \div (7^3)^2$$

$$(7^4 \times 7^3) \div 7^6 \Rightarrow 7^{(4+3-6)} \Rightarrow 7^1 = 7$$

$$9^3 \times 3^4 \div 27^2 = ?$$

- (A) 27 (B) 729 (C) 243 (D) 9 ☒ (E) 81

$$(3^2)^3 \times (3)^4 \div (3^3)^2$$

$$= 3^6 \times 3^4 \div 3^6$$

$$= 3^{(6+4-6)} \Rightarrow 3^4 \Rightarrow 81$$

$$(32^2 \times 8^3) \div 16^4 = ?$$

- (A) 4 (B) 16 ☒ (C) 8 (D) 32 (E) 64

$$[(2^5)^2 \times (2^3)^3] \div (2^4)^4$$

$$= (2^{10} \times 2^9) \div 2^{16}$$

$$= 2^{(10+9-16)} \Rightarrow 2^3 = 8$$

$$(40)^2 \div 25 \times 2 = 2^?$$

- (A) 3 (B) 5 (C) 6 ☒ (D) 7 (E) None of these

$$= \left(\frac{40 \times 40}{25} \right) \times 2$$

$$= 64 \times 2 \Rightarrow 2^6 \times 2^1 \Rightarrow 2^{(6+1)} \Rightarrow 2^7$$

$$(81 \times 7)^2 \div 3 = 49 \times 3^?$$

- (A) 8 (B) 7 (C) 10 (D) 5 (E) None of these

$$\Rightarrow \frac{81^2 \times 7^2}{3} = 7^2 \times 3^x$$

$$\Rightarrow \frac{(3^4)^2}{3^1} \Rightarrow 3^8 \div 3^1 \Rightarrow 3^{(8-1)} \Rightarrow 3^7 = 3^x$$

$$\boxed{x=7}$$

$$(2^{-3} + 2^{-4}) \div 2^{-5} = ?$$

- (A) 6 (B) 4 (C) 3 (D) 8 (E) 12

$$\frac{2^{-3} + 2^{-4}}{2^{-5}} \Rightarrow \frac{2^{-3}}{2^{-5}} + \frac{2^{-4}}{2^{-5}} \Rightarrow 2^{(-3)-(-5)} + 2^{(-4)-(-5)}$$

$$\Rightarrow (2^{-3+5}) + (2^{-4+5}) \Rightarrow 2^2 + 2^1 = 2^3 = 6$$

$$(81^2 \div 9^3) \times 3^2 = ?$$

- (A) 27 (B) 54 (C) 81 (D) 108 (E) 162

$$\Rightarrow [(3^4)^2 \div (3^2)^3] \times 3^2$$

$$\Rightarrow (3^8 \div 3^6) \times 3^2 \Rightarrow 3^{(8-6+2)} \Rightarrow 3^4 \Rightarrow 81$$

$$2^? = (8^5 \div 4^3)$$

- (A) 8 (B) 7 (C) 11 (D) 10 (E) 9

$$2^x = (2^3)^5 \div (2^2)^3$$

$$2^x = 2^{15} \div 2^6$$

$$2^x = 2^{(15-6)}$$

$$2^x = 2^9$$

$$\boxed{x=9}$$

$$? = (10^6 \div 2^6) \div 5^4$$

- (A) 5 (B) 10 (C) 50 (D) 25 (E) 125

$$\left(\frac{10^6}{2^6}\right) \div 5^4 \Rightarrow \left(\frac{10}{2}\right)^6 \div 5^4 \Rightarrow 5^6 \div 5^4$$

$$5^6 \div 5^4 \Rightarrow 5^{(6-4)} \Rightarrow 5^2 \Rightarrow 25$$

$$? = (4^7 \div 2^9) \times 8$$

- (A) 64 (B) 128 (C) 256 (D) 512 (E) 1024

$$\left(\frac{4^7}{2^9}\right) \times 8 \Rightarrow \frac{(2^2)^7}{2^9} \times (2)^3 \Rightarrow 2^{14} \div 2^9 \times 2^3$$

$$\Rightarrow 2^{(14-9+3)} \Rightarrow 2^8 = 256$$

If $3^{x+3} + 3^x = 756$, then $x = ?$

- (A) 1 (B) 4 (C) 2 (D) 3 (E) 5

$$3^{(x+3)} + 3^x = 756$$

$$(3^x \times 3^3) + 3^x = 756$$

$$(3^x \times 27) + 3^x = 756$$

$$3^x(27+1) = 756$$

$$3^x \times 28 = 756$$

$$3^x = 27$$

$$3^x = 3^3$$

$$x = 3$$

If $4^{x-2} = 8^{x-3}$, then $x = ?$

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

$$(2^2)^{x-2} = (2^3)^{x-3}$$

$$2^{2x-4} = 2^{3x-9}$$

$$2x-4 = 3x-9$$

$$x = 5$$

If $9^x = 27^{x-1}$, then $x = ?$

- (A) 3 (B) 2 (C) 1 (D) 4 (E) 5

$$(3^2)^x = (3^3)^{x-1}$$

$$3^{2x} = 3^{3x-3}$$

$$2x = 3x-3$$

$$x = 3$$

If $2^{x+3} = \frac{1}{8^{x-1}}$, then $x = ?$

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

$$2^{x+3} = 8^{-(x-1)}$$

$$2^{x+3} = 2^{3(-x+1)}$$

$$2^{x+3} = 2^{-3x+3}$$

$$x+3 = -3x+3$$

$$4x = 0$$

$$x = 0$$

$$(3^5 \times 9^2) \div 27^7 = 3^3$$

- (A) 3 (B) 5 (C) 1 (D) 2 (E) 4

$$(3^5 \times (3^2)^2) \div (3^3)^x = 3^3$$

$$3^{(5+4-3x)} = 3^3$$

$$9-3x=3 \Rightarrow 6=3x \Rightarrow x=2$$

$$(0.8)^7 = 0.512$$

- (A) 2 (B) 5 (C) 4 (D) 3 (E) 6

$$\left(\frac{8}{10}\right)^x = \left(\frac{512}{1000}\right)$$

$$\left(\frac{8}{10}\right)^x = \left(\frac{8}{10}\right)^3$$

$$x=3$$

$$\begin{cases} 512 = 8^3 \\ 1000 = 10^3 \end{cases}$$

$$32^3 \times 16^7 \div 8^4 = 2^{13}$$

- (A) 2 (B) 2.5 (C) 3 (D) 3.5 (E) 4

$$(2^5)^3 \times (2^4)^x \div (2^3)^4 = 2^{13}$$

$$2^{(15+4x-12)} = 2^{13}$$

$$3+4x=13 \Rightarrow 4x=10 \Rightarrow x=2.5$$

$$\text{If } 5^{x+2} = 125^{2x-1}, \text{ then } x = ?$$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

$$5^{x+2} = (5^3)^{2x-1}$$

$$5^{x+2} = 5^{3(2x-1)}$$

$$5^{x+2} = 5^{6x-3}$$

$$x+2=6x-3 \Rightarrow$$

$$5x=5$$

$$x=1$$

$$\text{If } 4^x + 4^{x-1} = 80, \text{ then } x = ?$$

- (A) 2 (B) 6 (C) 4 (D) 5 (E) 3

$$\left(4^x + \frac{4^x}{4}\right) = 80$$

$$4^x \left[1 + \frac{1}{4}\right] = 80$$

$$4^x \times \frac{5}{4} = 80$$

$$4^x = 64$$

$$4^x = 64$$

$$4^x = 4^3$$

$$x=3$$